

demonstration of effectiveness of each functionality. This project requires approximately 87,000 smart meters and a street-light automation solution that will prove substantial savings to the electricity department of the government of Puducherry (PED). CG's utility distribution franchise that operates in Jalgaon efficiently manages the multiple data obtained from the intelligent devices installed at all voltage levels, enhancing the quality of service provided to the utility's end users and drastically reduces technical and non-technical losses.

Authorities have come out with four types of electronic meters, and specifications for those have been released. Features of those are frozen on the payment side as well as the relay for remote connect/disconnect feature. "What is lacking is the communications protocol that is currently open ended. This is currently being managed by the system integrator based on what makes sense to them depending on the area," says Thakurdesai. Government is starting 14 smart grid pilot projects across cities in India, to implement certain technology, and to understand this from an Indian deployment perspective.

Under the current 12th 5-year Plan, the government will roll out full smart grids in pilot project cities, as well as the infrastructure for AMI to all consumers with load greater than 20kW. The government is also working on the implementation of dynamic tariffs, mandatory demand response programmes for certain categories of customers and a tariff mechanism for rooftop solar PVs.

We can also expect the first set of technical standards after completion of pilots, including standards for EVs and their charging infrastructure. Moreover, 10 per cent of utility technical personnel are to be trained in smart grid solutions.

"Now, there is an upcoming rule, which has already started in a few states, that people who are pumping harmonics into the network will be penalised. Harmonics is not gener-

India Smart Grid Task Force

The Ministry of Power established it in 2010 as an interdisciplinary group to explore how India can develop and deploy smart grid capabilities. Their deliberations and recommendations have covered a range of areas, from wide area monitoring systems for transmission, through renewable and distributed energy systems, automation and standards to cyber security. The task force collaborated closely with the Indian industry, through the India Smart Grid Forum, to develop this cross-functional understanding.

Do we really need smart meters to smarten our country's grid?

The smart grid will likely look different for different counties, and depends on many factors, relevant to the particular local operators, the geographic, the demographics and the government's agenda. The result should be a reliable network that provides affordable electricity to consumer. This is all what the customer really wants. The customer does not care how 'smart' the network is.

There have been scenarios where the government backed off on the roll out of smart meters after an outcry from the public who, to their big surprise, suddenly saw a new item listed on their electricity bill and were asked to pay up their contribution for its roll out.

Another hit against smart meters was when Attorney General of Connecticut warned the 2010 pilot results showed that smart meters had no beneficial impact on total energy usage or bill savings and that the advanced technology is very expensive. This was published in a statement from the office of the Attorney General of Connecticut.

A report, *Smart Metering—Ed Milliband's Poisoned Chalice* by Alex Henney, Ross Anderson, shows findings, including a study of smart metering in 11 countries, and suggests that smart meter technology offers little to no cost or energy savings.

Some important building blocks to be deployed in India

- Supervisory control and data acquisition (SCADA) systems with energy management and distribution management
- Information and communication technology to cover all substations and field offices with reliable communications solutions
- Graphical information systems which provide mapping of electrical network assets on maps
- Modernisation of substations with modern switchgear, numerical relays and modern test equipment
- Advanced metering infrastructure (AMI) and integrated distribution and substation automation
- Forecasting, dispatch and settlement tools
- Analytics to convert gathered data into business intelligence

ated from the source side but the load side. The consumer is the culprit who is using a lot of equipment generating harmonics. Now everyone will wake up," says A.V. Srinivasan, area manager—south, MB Controls and Systems. Big companies already have a solution to suppress harmonics; they are not allowing the harmonics to go out but are retaining them, which is also bad in a way.

Challenges faced in India

Developed countries are characterised by already having a reliable electrical infrastructure coupled with minuscule growth rates which allow them to eas-

ily focus on smartening up their grids with data gathering and analytics. However, in India, there needs to be a simultaneous focus on growing the basic electrical network to ensure that power is available to more remote villages, whilst at the same time upgrade and automate existing networks with smart grid technologies. What are the challenges in doing this?

Aging infrastructure. A large part of the infrastructure in India is several decades old, and reaching the end of their useful life.

The biggest challenge is the infrastructure in rural areas, where the range and distance are very large to the